

Francis Umo

Data Science Engineer 3 - Carbon Analytics & Validation Lead

GitHub: @franchise

Access Level: Maintainer

Reports To: @Goldokpa (Project Owner)

Project Duration: 3 Months

How This Role Fits You

With 8+ years of progressive experience in data analysis and business intelligence, you bring the deepest analytical maturity on this team. While others focus on building models and pipelines, you are the person who makes sure the numbers tell the right story and that the results are trustworthy.

Your expertise in Python, PostgreSQL, and SQL means you can build the carbon estimation models that require heavy data querying, aggregation, and statistical analysis. At Dataleum, you conducted data quality checks, developed dashboards to monitor financial data, and created reports that reduced fraud by 80% - that same rigour is exactly what's needed when validating whether our ML models are correctly estimating carbon loss from deforestation.

Your proficiency in Tableau and Power BI is a direct match for building the impact reporting layer. ClimateVision needs to produce clear, visual reports that conservation organizations and government agencies can act on. Your data storytelling background makes you the ideal person to translate raw model outputs into actionable intelligence.

Your cross-functional collaboration experience - working with IT teams, stakeholders, and bringing analytical models into production - means you understand how to bridge the gap between a data science experiment and a production metric that decision-makers rely on.

Your Role on ClimateVision

You own the analytics and validation layer - everything that turns raw model predictions into meaningful environmental metrics. If the ML model says 'this pixel is deforested,' you quantify what that means in tons of carbon, hectares of forest, and dollars of environmental impact.

Core Responsibilities

- Develop carbon stock estimation models (Random Forest, XGBoost regression)
- Build biomass-to-carbon conversion pipelines using allometric equations
- Implement uncertainty quantification (bootstrap, Monte Carlo, confidence intervals)
- Create ground truth validation framework - compare model outputs to known data
- Build statistical testing suite (hypothesis testing, A/B testing for model versions)
- Design and generate impact reports (area deforested, carbon lost, trends over time)
- Develop KPI dashboards for monitoring model performance and environmental outcomes
- Create validation notebooks demonstrating model accuracy across regions

Your Codebase Ownership

You are the primary owner of the following files and directories:

src/climatevision/analytics/	# PRIMARY OWNER - New analytics module
carbon.py	# Carbon stock estimation models
statistics.py	# Statistical testing & analysis
reporting.py	# Impact report generation
validation.py	# Ground truth validation framework
__init__.py	# Module exports
src/climatevision/models/	
regression.py	# PRIMARY OWNER - Biomass/carbon regression
src/climatevision/utils/	
metrics.py	# CO-OWNER - Extend with carbon metrics
notebooks/	
03_carbon_analysis.ipynb	# Carbon estimation analysis
04_model_validation.ipynb	# Validation & benchmarking
05_impact_reporting.ipynb	# Reporting notebook
outputs/	
reports/	# Generated impact reports
dashboards/	# Dashboard configs

Your 3-Month Delivery Timeline

MONTH 1: Foundation (Weeks 1-4)

Week 1-2: Carbon Estimation Models

- Research allometric equations for biomass estimation by forest type
- Build carbon.py - Random Forest & XGBoost regression for biomass prediction
- Create feature pipeline: spectral indices -> biomass -> carbon conversion
- Implement metrics for regression evaluation (RMSE, MAE, R-squared)

Week 3-4: Validation Framework

- Build validation.py - compare model predictions to ground truth datasets
- Source and integrate reference data (Global Forest Watch, forest inventory data)
- Create confusion matrix, precision/recall analysis for segmentation outputs
- Write 04_model_validation.ipynb with baseline validation results

MONTH 2: Advanced Analytics (Weeks 5-8)

Week 5-6: Uncertainty & Statistical Testing

- Implement bootstrap confidence intervals for carbon estimates

- Build Monte Carlo simulation for uncertainty propagation
- Create statistics.py - hypothesis testing, trend analysis functions
- Implement A/B testing framework for comparing model versions

Week 7-8: Impact Reporting

- Build reporting.py - automated report generation (PDF/HTML)
- Design KPI framework: hectares lost, carbon tons, trend direction
- Create 05_impact_reporting.ipynb - template for regional impact reports
- Integrate with PostgreSQL for historical metric storage

MONTH 3: Production Readiness (Weeks 9-12)

Week 9-10: Dashboard & Integration

- Build dashboard data endpoints (feed metrics to frontend charts)
- Create time-series analysis for deforestation trend tracking
- Implement anomaly detection for unusual forest loss patterns

Week 11-12: Documentation & Case Studies

- Produce 3 regional case study reports (Amazon, Congo, Southeast Asia)
- Write comprehensive documentation for analytics module
- Final validation sweep across all model outputs
- Performance benchmarking and accuracy documentation

Your Git Workflow

```
# Create feature branches from develop
git checkout develop
git pull origin develop
git checkout -b feature/analytics-carbon-estimation

# Your branch naming convention:
feature/analytics-*      (new analytics features)
fix/analytics-*          (bug fixes)
refactor/analytics-*     (code restructuring)
```

All PRs go to the develop branch. PRs require at least 1 review. Tag @edoh-Onuh for reviews on model evaluation metrics, and @Oshgig for data format questions.

Your Key Collaborators

- @edoh-Onuh (ML Lead) - Their model predictions are your primary input. Coordinate on output formats, probability thresholds, and confidence scores.
- @Oshgig (Data Pipeline Lead) - She provides the preprocessed data you need for carbon regression features. Align on spectral indices and metadata.
- Olufemi Taiwo (API Lead) - Your analytics endpoints need to be exposed through the API. Coordinate on response

schemas.

- @cutewizzy11 (Full-Stack) - Frontend dashboards visualize your metrics. Provide JSON data contracts for charts.

Your Code Pipeline

Your pipeline starts where the ML model ends - taking prediction masks and turning them into carbon impact numbers and stakeholder reports.

Step 1: Environment Setup

```
git clone https://github.com/Climate-Vision/ClimateVision.git
cd ClimateVision
pip install -r requirements.txt

# Verify analytics dependencies
python -c "import xgboost, sklearn, mlflow, optuna; print('Analytics stack ready')"
```

Step 2: Run Inference to Get Prediction Masks

```
# Generate deforestation masks from a trained model
python scripts/infer.py \
  --bbox "-60,-15,-45,5" \
  --date 2023-06-01 \
  --analysis_type deforestation \
  --output outputs/masks/

# Output: outputs/masks/deforestation_mask.tif + confidence_scores.npy
```

Step 3: Estimate Carbon Loss

```
# Run carbon stock estimation on the prediction mask
python - <<'EOF'
from climatevision.analytics.carbon import estimate_carbon
result = estimate_carbon(
    mask_path='outputs/masks/deforestation_mask.tif',
    region='amazon',
    forest_type='tropical_moist'
)
print(f"Deforested area: {result['hectares']:.1f} ha")
print(f"Carbon lost:      {result['carbon_tonnes']:.1f} tCO2e")
print(f"Confidence CI:   {result['ci_lower']:.1f} - {result['ci_upper']:.1f} tCO2e")
EOF
```

Step 4: Validate Against Ground Truth

```
# Compare model outputs to Global Forest Watch reference data
python - <<'EOF'
from climatevision.analytics.validation import validate_predictions
metrics = validate_predictions(
    pred_mask='outputs/masks/deforestation_mask.tif',
    ground_truth='data/ground_truth/amazon_gfw_2023.tif'
)
```

```
print(f"IoU: {metrics['iou']:.3f} | F1: {metrics['f1']:.3f} | Precision: {metrics['precision']:.3f}")
EOF
```

Step 5: Generate Impact Report

```
# Auto-generate a PDF/HTML impact report for stakeholders
python - <<'EOF'
from climatevision.analytics.reporting import generate_report
generate_report(
    region='amazon',
    period='2023-Q2',
    carbon_result=result,
    validation_metrics=metrics,
    output_dir='outputs/reports/'
)
EOF

# Output: outputs/reports/amazon_2023-Q2_impact_report.pdf
```

Step 6: Commit & Push Your Work

```
# Switch to your git identity
source team_docs/switch_user.sh francis

git checkout develop && git pull origin develop
git checkout -b feature/analytics-carbon-estimation

git add src/climatevision/analytics/
git add notebooks/03_carbon_analysis.ipynb
git commit -m "feat(analytics): add carbon stock estimation with confidence intervals"

git push francis feature/analytics-carbon-estimation
```